Lagrange Multipliers and Their Application in Economics: Cost Minimisation

Undergraduate Report

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1. Introduction

When you search about the most popular methods for constrained optimization, the Lagrange multiplier method will come up. Joseph-Louis Lagrange created the Lagrange multiplier methods in the late 1700s[1], and it is still used in mathematical economics, until this day. The key part is that it converts a constrained problem into an unconstrained one by introducing an auxiliary variable, the Lagrange multiplier, and combining the objective function and the constraint into a single expression called the Lagrangian. Then the partial derivative is taken and set to zero and after we get a system of equations from which we get the optimal values

Constrained optimization problems like this can be found in economics, where consumers allocate their income across goods to maximise utility, and firms choose input combinations to minimise the cost of producing a given level of output. And in both cases, the decision maker has an objective function and faces a constraint that must be satisfied. The method of Lagrange multipliers provides a systematic way to solve these problems, and the multiplier itself has a direct economic meaning. At the optimum, it measures how much the value of the objective function would change if the constraint were relaxed by one unit. In consumer theory, this corresponds to the marginal utility of income. In the firm's cost minimisation problem, it gives the marginal cost of output. For this reason, the multiplier is often referred to as the shadow price of the constraint.

Our report explores both the mathematical foundations and the economic applications of the method. Section 2 sets up the general constrained optimisation problem, introduces the Lagrangian, derives the first-order necessary conditions, and discusses the geometric interpretation, specifically, that at the optimum, the level curve of the objective function is tangent to the constraint set. We also briefly cover the second-order sufficiency conditions using the bordered Hessian and the extension to problems with multiple constraints.

Section 3 turns to economic applications. We first consider consumer utility maximisation, where a consumer with a square-root utility function chooses quantities of two goods subject to a budget constraint. We derive the optimality condition that the marginal rate of substitution must equal the price ratio and solve a numerical example. We then set up the firm's cost minimisation problem, where a competitive firm uses capital and labour to produce a target output level at minimum cost. So the optimality condition requires the marginal rate of technical substitution to equal the ratio of input prices, which means the last dollar spent on each input must yield the same marginal product.

Section 4 discusses the economic interpretation of the Lagrange multiplier in more detail, connecting it to the envelope theorem and showing that the multiplier equals the derivative of the optimal cost with respect to the output target. Section 5 provides a complete

worked example using Cobb–Douglas production technology. We derive the conditional factor demand functions, obtain the minimum cost function in closed form, and present a numerical illustration with specific parameter values. The example shows how the theoretical results translate into concrete numbers in particular, we compute the shadow price and interpret it as the cost of producing one additional unit of output.

2. Mathematical Foundations

2.1. Constrained Optimisation — Setup

Let $f : \mathbb{R}^n \to \mathbb{R}$ be the objective function and $g : \mathbb{R}^n \to \mathbb{R}$ the equality constraint. The general single-constraint problem is:

$$\min_{\mathbf{x}\in\mathbb{R}^n} f(\mathbf{x}) \quad \text{subject to} \quad g(\mathbf{x}) = c, \tag{1}$$

where $c \in \mathbb{R}$ is a given constant.

2.2. The Lagrangian

Definition 2.1 (Lagrangian function)**.** *The* Lagrangian *associated with problem* (1) *is*

$$\mathcal{L}(\mathbf{x}, \lambda) \;=\; f(\mathbf{x}) - \lambda \Big[g(\mathbf{x}) - c\Big], \tag{2}$$

where $\lambda \in \mathbb{R}$ *is the* Lagrange multiplier.

Remark 2.2. *Some texts write* $\mathcal{L} = f + \lambda(g - c)$*. The sign convention in* (2) *is standard in economics and ensures* $\lambda > 0$ *at a minimum when the constraint is binding and* f, g *are well-behaved.*

2.3. First-Order Necessary Conditions

Theorem 2.3 (Lagrange's Theorem)**.** *Let* f *and* g *be continuously differentiable. If* $\mathbf{x}^*$ *is a local solution of* (1) *and the constraint qualification* $\nabla g(\mathbf{x}^*) \neq \mathbf{0}$ *holds, then there exists* $\lambda^* \in \mathbb{R}$ *such that*

$$\nabla_{\mathbf{x}}\, \mathcal{L}(\mathbf{x}^*, \lambda^*) = \nabla f(\mathbf{x}^*) - \lambda^*\, \nabla g(\mathbf{x}^*) = \mathbf{0}, \tag{3}$$

$$\frac{\partial \mathcal{L}}{\partial \lambda}(\mathbf{x}^*, \lambda^*) = g(\mathbf{x}^*) - c = 0. \tag{4}$$

In component form, for $i = 1, \ldots, n$:

$$\frac{\partial f}{\partial x_i}(\mathbf{x}^*) = \lambda^*\, \frac{\partial g}{\partial x_i}(\mathbf{x}^*). \tag{5}$$

Together with the constraint equation, this yields $n + 1$ equations in $n + 1$ unknowns

$(x_1^*, \ldots, x_n^*, \lambda^*)$.

2.4. Geometric Interpretation

Equation (3) states that at the optimum the gradient of f is *parallel* to the gradient of g. Geometrically, the level curve of f is *tangent* to the constraint set $\{g(\mathbf{x}) = c\}$. Moving along the constraint can no longer improve f once this tangency holds.

2.5. Second-Order Sufficiency Conditions

The first-order conditions are necessary but not sufficient in general. To confirm a *minimum*, one examines the *bordered Hessian*:

$$\bar{H} = \begin{pmatrix} 0 & \nabla g^\top \\ \nabla g & \nabla^2_{xx}\mathcal{L} \end{pmatrix}. \tag{6}$$

For a two-variable problem ($n = 2$), the bordered Hessian is 3×3 and the second-order sufficient condition for a local *minimum* is that its leading 3×3 principal minor satisfies $\det(\bar{H}) > 0$.

2.6. Multiple Constraints

For $m < n$ constraints $g_j(\mathbf{x}) = c_j$, $j = 1, \ldots, m$, the Lagrangian generalises to

$$\mathcal{L}(\mathbf{x}, \boldsymbol{\lambda}) = f(\mathbf{x}) - \sum_{j=1}^{m} \lambda_j \Big[g_j(\mathbf{x}) - c_j\Big], \tag{7}$$

and the first-order conditions become $\nabla_{\mathbf{x}}\mathcal{L} = \mathbf{0}$ and $g_j(\mathbf{x}) = c_j$ for all j.

3. Consumer Utility Maximisation

One of the most common applications of constrained optimisation in economics is the problem of utility maximisation. For instance, consumers choose quantities of goods in order to maximise their utility (assuming them to be rational!) subject to a budget constraint.

This could also be applied in other domains as well, such as Labour-leisure choice, where workers decide how many hours to work versus how much free time to enjoy. In theory, workers choose the point where the utility gained from extra income is equal to the utility lost from reduced leisure. Suffice to say, there are many applications of utility maximisation across economic analysis.

3.1. Utility Function

Assume that a consumer derives utility from two goods, x and y. The consumer's preferences are represented by the following utility function:

$$U(x, y) = \sqrt{x}\sqrt{y} \tag{8}$$

For our purposes, we assume this to be in square root form, as it exhibits diminishing marginal utility and implies that the consumer values both goods.

However, in principle we can represent it in different forms. For example, it can be a linear utility function with constant marginal utility, where a and b represent marginal utility derived from each good:

$$U(x, y) = ax + by \tag{9}$$

In most economic models, utility functions are assumed to have diminishing marginal utility. This is often represented by a Cobb–Douglas utility function, where the exponents α and β are less than one[5]:

$$U(x, y) = x^{\alpha}y^{\beta} \tag{10}$$

3.2. Budget Constraint

The consumer faces a budget constraint determined by prices and their income. Let p_x and p_y denote the prices of goods x and y, and let I denote income. The budget constraint is given by

$$p_x x + p_y y = I \tag{11}$$

This equation represents all combinations of x and y that the consumer can afford. Here we assume equality to find the optimal point, as if they spend less than their income, they can buy more goods to increase their utility.

3.3. The Lagrangian

To maximise utility subject to the budget constraint, we construct the Lagrangian function[3]:

$$\mathcal{L}(x, y, \lambda) = \sqrt{x}\sqrt{y} + \lambda\left(I - p_x x - p_y y\right) \tag{12}$$

The first part of the equation represents the consumer's utility from consuming quantities of good x and y, while the second term incorporates the budget constraint into the optimisation problem by representing the consumer's remaining income after spending $p_x x$ on good x and $p_y y$ on good y.

The parameter λ is the Lagrange multiplier associated with the budget constraint. It represents the marginal value of relaxing the constraint, that is, how much the maximum utility of the consumer would increase if the consumer had one additional unit of income. For this reason, it is often interpreted as the marginal utility of income.

If the λ value is large, extra income would increase utility considerably. However, if the λ value is small, extra income would not increase utility greatly.

3.4. First-Order Conditions

To find the optimal choice of x and y, we take the partial derivatives of the Lagrangian and set them equal to zero:

$$\frac{\partial \mathcal{L}}{\partial x} = \frac{1}{2}\frac{\sqrt{y}}{\sqrt{x}} - \lambda p_x = 0 \tag{13}$$

$$\frac{\partial \mathcal{L}}{\partial y} = \frac{1}{2}\frac{\sqrt{x}}{\sqrt{y}} - \lambda p_y = 0 \tag{14}$$

$$\frac{\partial \mathcal{L}}{\partial \lambda} = I - p_x x - p_y y = 0 \tag{15}$$

The third condition simply reproduces the budget constraint.

3.5. Optimality Condition

Dividing the first two equations yields

$$\frac{\sqrt{y}/\sqrt{x}}{\sqrt{x}/\sqrt{y}} = \frac{p_x}{p_y} \tag{16}$$

This simplifies to

$$\frac{y}{x} = \frac{p_x}{p_y} \tag{17}$$

This condition implies that the marginal rate of substitution equals the ratio of prices [2]:

$$MRS = \frac{p_x}{p_y} \tag{18}$$

3.6. Economic Interpretation of the Lagrange Multiplier

The Lagrange multiplier λ represents the marginal utility of income. It measures how much utility would increase if the consumer had one additional unit of income. For this reason, λ is often referred to as the *shadow price* of the budget constraint [2].

3.7. Numerical Example

To illustrate the solution numerically, consider the consumer's optimisation problem

$$\max_{x,y} U(x, y) = \sqrt{x}\sqrt{y} \tag{19}$$

subject to the budget constraint:

$$2x + 4y = 100. \tag{20}$$

Using the optimality condition derived above, we solve for the optimal consumption bundle.

Suppose the consumer has income $I = 100$, the price of the good x is $p_x = 2$, and the price of the good y is $p_y = 4$. We now solve for the optimal bundle step by step.

From equation (17) we know that

$$\frac{y}{x} = \frac{2}{4} = \frac{1}{2} \tag{21}$$

So,

$$y = \frac{1}{2}x \tag{22}$$

Next, substitute this into the budget constraint (20):

Substituting equation (22) into equation (20) gives

$$2x + 4\left(\frac{1}{2}x\right) = 100 \tag{23}$$

Simplifying:

$$2x + 2x = 100 \tag{24}$$

$$4x = 100 \tag{25}$$

$$x^* = 25 \tag{26}$$

Now substitute $x^* = 25$ into equation (22):

$$y^* = \frac{1}{2}(25) = 12.5 \tag{27}$$

Hence, the optimal consumption bundle is

$$x^* = 25, \qquad y^* = 12.5 \tag{28}$$

At this point the consumer spends all income and achieves the maximum attainable utility given the budget constraint.

3.8. The Firm's Cost Minimisation Problem

A competitive firm produces a single output q using two inputs, capital K and labour L, with prices $r > 0$ (rental rate) and $w > 0$ (wage rate) respectively. The technology is summarised by a production function $F(K, L)$.

The firm wishes to produce at least $\bar{q}$ units of output at minimum cost:

$$\min_{K,\, L \geq 0} \; C(K, L) = rK + wL \quad \text{subject to} \quad F(K, L) = \bar{q}. \tag{29}$$

Here the objective is the *total cost* function C and the constraint is the *isoquant* $F(K, L) = \bar{q}$.

3.9. Lagrangian Formulation

The Lagrangian is

$$\mathcal{L}(K, L, \lambda) = rK + wL - \lambda\left[F(K, L) - \bar{q}\right]. \tag{30}$$

3.10. First-Order Conditions

Setting partial derivatives to zero:

$$\frac{\partial \mathcal{L}}{\partial K} = r - \lambda F_K = 0 \implies \lambda = \frac{r}{F_K}, \tag{31}$$

$$\frac{\partial \mathcal{L}}{\partial L} = w - \lambda F_L = 0 \implies \lambda = \frac{w}{F_L}, \tag{32}$$

$$\frac{\partial \mathcal{L}}{\partial \lambda} = -(F(K, L) - \bar{q}) = 0. \tag{33}$$

where $F_K = \partial F/\partial K$ and $F_L = \partial F/\partial L$ are the *marginal products* of capital and labour.

3.11. The Optimality Condition: Equalising Marginal Products per Dollar

Equating the two expressions for λ from (31) and (32):

$$\frac{F_K}{r} = \frac{F_L}{w} \quad \iff \quad \frac{F_K}{F_L} = \frac{r}{w}. \tag{34}$$

The left-hand side is the *Marginal Rate of Technical Substitution* (MRTS), which measures how much capital can replace one unit of labour while keeping output constant. The right-hand side is the *input price ratio*.

Interpretation: Condition (34) states that at the optimum, the slope of the isoquant equals the slope of the isocost line. Equivalently, the last dollar spent on capital and the last dollar spent on labour must yield the same marginal product. If this were not so, the firm could reallocate expenditure toward the cheaper input per unit of output and reduce costs while maintaining $\bar{q}$ [7].

4. Economic Interpretation of the Lagrange Multiplier

The multiplier λ^* has a natural and important economic meaning. Differentiating the minimum cost $C^* = C^*(r, w, \bar{q})$ with respect to the output target $\bar{q}$, one can show (via the envelope theorem) that:[4].:

$$\lambda^* = \frac{\partial C^*}{\partial \bar{q}}. \tag{35}$$

The Lagrange multiplier equals the firm's marginal cost [2]. — the additional expenditure required to produce one more unit of output at the optimum. For this reason λ is also called the *shadow price* of the constraint: it measures how much the objective (cost) would increase if the right-hand side of the constraint (target output) were relaxed by one unit.

This is a profound result: a scalar that appeared as a mathematical device to enforce the

constraint turns out to carry direct economic information about the value of relaxing that constraint.[6].

5. Worked Example: Cobb–Douglas Technology

5.1. Setup

Suppose the production function is Cobb–Douglas[5]:

$$F(K, L) = K^{\alpha} L^{\beta}, \qquad \alpha, \beta > 0,\ \alpha + \beta = 1. \tag{36}$$

The firm minimises cost subject to producing $\bar{q}$ units:

$$\min_{K,L}\ rK + wL \quad \text{s.t.} \quad K^{\alpha} L^{\beta} = \bar{q}. \tag{37}$$

5.2. Applying the Lagrange Conditions

The Lagrangian is

$$\mathcal{L}(K, L, \lambda) = rK + wL - \lambda\,(K^{\alpha} L^{\beta} - \bar{q}).$$

First-order conditions:

$$r = \lambda\,\alpha\,K^{\alpha-1} L^{\beta}, \tag{38}$$

$$w = \lambda\,\beta\,K^{\alpha} L^{\beta-1}, \tag{39}$$

$$K^{\alpha} L^{\beta} = \bar{q}. \tag{40}$$

Dividing (38) by (39):

$$\frac{r}{w} = \frac{\alpha}{\beta} \cdot \frac{L}{K} \quad \Longrightarrow \quad K = \frac{\alpha}{\beta} \cdot \frac{w}{r}\,L. \tag{41}$$

5.3. Conditional Factor Demands

Substituting (41) into the constraint (40):

$$\left(\frac{\alpha}{\beta} \cdot \frac{w}{r}\right)^{\alpha} L^{\alpha+\beta} = \bar{q}$$

$$L^{*} = \bar{q}\left(\frac{\alpha w}{\beta r}\right)^{-\alpha} = \bar{q}\left(\frac{\beta r}{\alpha w}\right)^{\alpha}, \tag{42}$$

and similarly:

$$K^{*} = \bar{q}\left(\frac{\alpha w}{\beta r}\right)^{\beta}. \tag{43}$$

These are the *conditional factor demand* functions; they give optimal input use as a function of prices and the output target.

5.4. Minimum Cost Function

Substituting K^* and L^* back into the cost function:

$$C^*(r, w, \bar{q}) = rK^* + wL^* = \bar{q}\left(\frac{r}{\alpha}\right)^{\alpha}\left(\frac{w}{\beta}\right)^{\beta}(\alpha^{\alpha}\beta^{\beta})^{-1} \cdot \alpha^{\alpha}\beta^{\beta}. \tag{44}$$

Simplifying (using $\alpha + \beta = 1$):

$$\boxed{C^*(r, w, \bar{q}) = \bar{q}\left(\frac{r}{\alpha}\right)^{\alpha}\left(\frac{w}{\beta}\right)^{\beta}.} \tag{45}$$

5.5. Numerical Illustration

Set $\alpha = 1/3$, $\beta = 2/3$, $r = 3$, $w = 6$, $\bar{q} = 100$.

Optimal inputs:

$$K^* = 100 \cdot \left(\frac{(1/3) \cdot 6}{(2/3) \cdot 3}\right)^{2/3} = 100 \cdot (1)^{2/3} = 100,$$

$$L^* = 100 \cdot \left(\frac{(2/3) \cdot 3}{(1/3) \cdot 6}\right)^{1/3} = 100 \cdot (1)^{1/3} = 100.$$

Minimum cost:

$$C^* = 3(100) + 6(100) = 300 + 600 = 900.$$

Shadow price (marginal cost):

$$\lambda^* = \frac{\partial C^*}{\partial \bar{q}} = \left(\frac{r}{\alpha}\right)^{\alpha}\left(\frac{w}{\beta}\right)^{\beta} = 9^{1/3} \cdot 9^{2/3} = 9.$$

Interpretation: producing one additional unit beyond $\bar{q} = 100$ costs the firm approximately \$9.

Table 1: Summary of the Numerical Example

Quantity	Symbol	Value
Output target	$\bar{q}$	100
Rental rate	r	3
Wage rate	w	6
Optimal capital	K^*	100
Optimal labour	L^*	100
Minimum cost	C^*	900
Marginal cost (shadow price)	λ^*	9

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